thylinao at ChildSafeAds 2026: When the Divisor Is Part of the Hidden Test Set

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Abstract

ChildSafeAds scores systems with macro-F1 averaged over the labels *present* in each scored sample. This paper shows that the convention makes the divisor of every reported column a property of the hidden gold rather than of the label space, and reports three measured consequences. First, the divisor is the dominant source of variance in a reported score: on the out-of-fold split, whether a two-instance class is drawn moves the reported mean by 0.053 with predictions held bit-identical, about six times the gap between the top two entries. Second, the divisor and the gold count behind it are recoverable in closed form from the organisers' published constant-baseline row; the development split's counts are recovered exactly, and the evaluation split is predicted to contain 235 `physical_goods` instances, which the overview paper's Table 3 reports as 235. Third, an ablation over the four data access levels shows that the channel name adds nothing (-0.003, CI $[-0.012, +0.005]$) and that the outbound page crawl adds product identity but not compliance (ST3 across all four levels $+0.024$, CI spanning zero), while the compliance labels themselves agree across the organisers' two annotator models on only 44.4% of instances. The system described here scores 0.6537. The entry submitted returned 0.7260 but contained labels the system did not produce, which the task Terms prohibit. That was disclosed to the organisers and is documented in Section 9. No rank is claimed, for reasons the noise measurement makes quantitative.

1 Introduction

ChildSafeAds (Bertaglia et al., 2026) asks three questions about sponsored segments in YouTube videos likely to reach children: what kind of offer is promoted (ST1, five classes, single label), which product categories apply (ST2, twelve, multi-label), and which advertising-law concerns the segment raises (ST3, eight flags, multi-label). It follows a line of computational work on disclosure practice in influencer marketing (Mathur et al., 2018; Bertaglia et al., 2025) and on commercial pressure directed at children (De Veirman et al., 2019; Moti et al., 2024). Systems are ranked by the mean of the three macro-F1 columns; twenty-one teams finished the evaluation phase on Codabench (Xu et al., 2022).

The system entered is a conventional one: six members, comprising two fine-tuned encoders, a zero-shot prompted 32B model, a LoRA-tuned 14B classifier, a TF-IDF baseline and a level-2 stacker, behind a decision layer whose thresholds are fitted only on channel-grouped out-of-fold predictions. Most of this paper concerns the evaluation rather than the system, because the attempt to establish whether individual changes to the system were real led to measurements about the metric that seem more broadly useful.

Contributions. **(i)** A measurement showing that under present-label-set macro-F1 the metric's divisor, rather than the system, is the largest single source of variance in a reported score (§5). **(ii)** A closed form recovering the divisor and the corresponding hidden gold count from a published constant-baseline row, validated where gold is known and confirmed against the overview paper on the evaluation split (§6). **(iii)** An answer to the question the task was built around, namely what each data access level provides, with the finding that one level adds nothing measurable and that the most expensive one adds nothing to the compliance task (§7). **(iv)** A quantification of how much of the benchmark is reachable by advertiser memorisation, a property the organisers document but do not size (§8).

Disclosure. The predictions submitted as the final entry contained three labels that were set by hand rather than produced by the system, and an earlier submitted entry contained a fourth. The competition Terms state, under Competition Conduct, item 6: *"Do not manually label the test set. Predictions on the evaluation phase must be produced by your system."* This was raised with the organisers after the phase closed and before any external party had identified it. Section 9 gives the full account. The score reported throughout as the system's is 0.6537, the returned score of the last entry containing no hand-set label. No rank is claimed anywhere in this paper.

2 Task, data, and a metric recovered empirically

The dataset pairs a SponsorBlock-marked segment with its transcript, video and channel context, and a sales page linked from the description, across 3,360 videos from 939 channels (Bertaglia et al., 2026). Labels were produced by GPT-5.4 under expert review of the taxonomy and prompts, with a second model labelling the development split independently, a protocol now common in dataset construction (Gilardi et al., 2023). Splits are disjoint by channel; as the overview's Table 1 caption states, *"Brands and destinations are not held out."* Section 8 returns to that.

The scoring convention is not specified. The overview defines the metric as *"F1 for every label and average the labels equally (macro-F1)"*. That does not fix the label space over which the average is taken, and the choice is consequential: a class absent from both gold and predictions may be dropped from the average, or scored as 0, or as 1. Which is intended is a known source of ambiguity in reported macro-F1 (Opitz and Burst, 2019; Opitz, 2024). The convention was recovered empirically, by scoring the organisers' constant-class baseline, whose per-column results are published on the leaderboard, under a 2×2 grid of label-set and zero-division conventions. Only the *present-label-set* convention, which averages over classes appearing in the gold or the predictions of the scored rows, reproduces the published row (ST1 0.150970 against a published 0.1510; a fixed five-class average gives 0.120776). This is the default path through `sklearn`'s `f1_score` when `labels` is left unset (Pedregosa et al., 2011), so it is what a scoring script inherits by default rather than by deliberate choice. Everything below uses the pinned convention.

Under that convention the divisor of each column is a function of the sample being scored. That property produces the two results in §5 and §6.

3 System

Six members write per-split class-probability matrices, which a decision layer converts to labels. Table 1 gives the roster, its compute, and each member's measured contribution.

The two encoders (DeBERTa-v3-large (He et al., 2021), ModernBERT-large (Warner et al., 2025)) are fine-tuned with three seeds each over a 5-fold channel-grouped split, with per-field token budgets and field dropout. The zero-shot lane prompts Qwen3-32B (Qwen Team, 2025) through vLLM (Kwon et al., 2023) with the task taxonomy quoted verbatim, guided JSON decoding, and temperature 0; a data-guard instruction directs the model to treat the instance, which contains imperatives and discount codes, as data rather than as instructions. The LoRA lane (Hu et al., 2022) fine-tunes Qwen3-14B as a sequence classifier and is retained for ST2 only, the one column where its gain had an interval excluding zero. A level-2 logistic stacker (Wolpert, 1992) over fold-honest member predictions supplies ST3.

The decision layer applies a prior-corrected argmax on ST1, per-class thresholds fitted by grid search on pooled out-of-fold predictions with a low-support plug-in fallback and a per-fold stability floor, and three ST3 hard rules that zero $p(\texttt{undisclosed\_advertising})$ where a disclosure demonstrably exists: the platform's paid-promotion label, a sponsorship term spoken in the transcript, or a sponsorship phrase in the description. Those rules fire on 2,321 of the 2,857 labelled rows, with a single gold violation among them. Folds, thresholds and every decision parameter are fitted on out-of-fold predictions only; the development split was held as a direction check.

4 Results

Table 2 gives the reference numbers. Uncertainty throughout is a channel-cluster bootstrap (Efron, 1979): channels are resampled with replacement, decisions are emitted once so that what is measured is the sampling noise of the estimate rather than tuning noise, and paired comparisons score both arms on the identical draw.

The scale of that noise is worth reporting first. Resampling 153 channels with replacement to evaluation size over $2{,}000$ draws, a 503-row scored estimate has a standard error of 0.0326 on the mean

Member	Model	GPU-h	Removal cost
m3q32A	Qwen3-32B, zero-shot	1.5	−0.0199 *
s3s	logistic stacker, ST3	n/a	−0.0098 *
m1m	ModernBERT-large	6	−0.0083 *
m2qS2	Qwen3-14B LoRA, ST2	7–8	−0.0072 *
m0s	TF-IDF, stacked	n/a	−0.0035
m1d	DeBERTa-v3-large	6	−0.0000

Table 1: Ensemble roster. Removal cost is the paired change in out-of-fold mean macro-F1 when the member is dropped and the whole decision layer refitted without it; a more negative value indicates a larger contribution. An asterisk marks a 95% interval excluding zero (4,000 channel resamples). GPU-hours are per member: the encoder lane spent about 12 hours over six runs, and the LoRA lane about 22 over three sub-task jobs, of which only the ST2 job is used here. The zero-shot member is the most valuable and among the cheapest; removing the fine-tuned DeBERTa, six GPU-hours of training, is indistinguishable from making no change.

	OOF	dev	evaluation split	
	(2353)	(504)	system	submitted
ST1	0.6369	0.8519	0.6021	0.8049
ST2	0.8423	0.7572	0.7761	0.7900
ST3	0.6199	0.5966	0.5830	0.5830
mean	0.6997	0.7352	**0.6537**	0.7260

Table 2: Held-out results. The *system* column is the last submitted entry containing no hand-set label, and is the figure reported throughout as the system's result. The *submitted* column is the returned score of the final entry, which contains three labels the system did not produce (§9).

(ST1 0.0869, ST2 0.0254, ST3 0.0446); drawing channel-disjoint subsamples without replacement instead gives 0.0327. The four adjacent gaps in the published top five are 0.0087, 0.0181, 0.0057 and 0.0323, so none of them reaches one standard error of the estimate that produced them. Evaluation sets of this size support weaker conclusions than they are commonly asked to support (Card et al., 2020; Dror et al., 2018). This paper therefore does not interpret its own position in that ordering.

5 The divisor is the dominant variance component

ST1 `other` has two gold instances in the 2,353 out-of-fold rows, spread over two of the 632 channels, and the system never predicts it. Under present-label-set semantics its presence in the gold alone admits it to the average, at F1 $= 0$.

Resampling channels, a gold `other` row is drawn

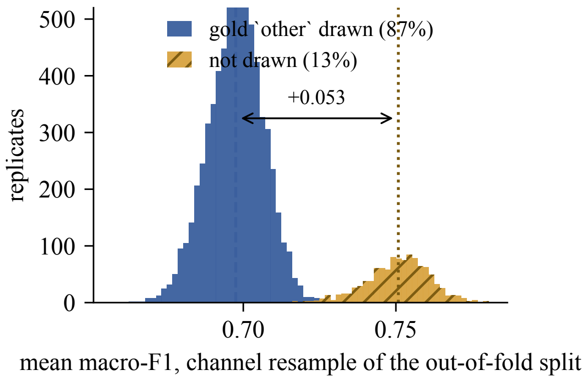

Figure 1: Bootstrap distribution of the system's reported mean macro-F1 over 8,000 channel resamples of the out-of-fold split. The predictions are identical in every replicate. The separation is due entirely to whether the resample contains one of the two gold `other` rows, which changes the ST1 average from five classes to four.

in 87.1% of replicates. Conditioning on that event partitions the bootstrap distribution into two nearly degenerate modes (Figure 1). With the class present, ST1 averages over five classes and the mean metric is 0.6976 (s.e. 0.0090). With it absent, ST1 averages over four and the mean metric is 0.7506 (s.e. 0.0102). The predictions are bit-identical in both regimes. The ST1 column moves by $+0.1604$, against a predicted $+0.1590$, since a four-class average is exactly $5/4$ of a five-class one. The reported mean moves by $+0.0530$.

Three consequences follow. First, a marginal standard error is not a sufficient summary of a score on this metric: the marginal figure of 0.0200 is a mixture across two regimes, roughly twice the within-regime figure, and none of the excess reflects a property of the system. Second, 0.0530 is about six times the 0.0087 separating the first and second entries on the final board, so two rows of a class no system emits carry more weight than the difference between the two leading systems. Third, whether to emit such a class is not decidable from the training data: firing and missing costs a false positive, while never firing costs the whole class slot, and which is worse depends on a property of the hidden gold. That decision was made incorrectly here, at a cost stated in §6.

6 The divisor is recoverable

Consider a predictor emitting one constant class c on all N instances, with g gold instances of c. Then $tp = g$, $fp = N - g$, $fn = 0$, so $\mathrm{F1}(c) = 2g/(N+g)$; every other gold-present class scores 0;

and classes absent from both gold and predictions are dropped. With K classes in the present set, the reported column is $S = 2g/\big(K(N+g)\big)$, which rearranges to

$$g = \frac{S\,K\,N}{2 - S\,K}. \quad (1)$$

S is published to four decimals and g must be an integer, so for each candidate K the equation either admits an integer solution reproducing the published rounding or it does not.

The organisers submitted a constant `physical_goods` baseline to both phases, which provides a control where the answer is known. On the development split, $S = 0.1510$ and $N = 504$: enumerating K, only $K = 4$ admits an integer solution, $g = 218$, which back-substitutes to 0.150970. The development gold contains exactly 218 `physical_goods` instances and exactly four of the five ST1 classes. On the evaluation split, $S = 0.1274$ and $N = 503$ leave two candidates within a five-class space: $K = 4$ with $g = 172$, and $K = 5$ with $g = 235$. They differ on the point at issue, because $K = 5$ saturates the ST1 label space and therefore requires `other`, the class this system suppresses, to be present. The pooled labelled data has a `physical_goods` rate of 0.4701; $235/503 = 0.467$ sits on that rate, while $172/503 = 0.342$ would require a shift of more than three standard deviations of a channel-block bootstrap of it. The overview paper's Table 3 reports the evaluation split's ST1 gold as 235 `physical_goods` and one `other`.

The relevance of the inversion here is what its absence cost. The system sets `other` never to fire, on the reasoning that a class with two training instances is more likely absent than present. That decision cost a full fifth of the ST1 column, four predicted classes divided by five present, which §9 measures at 0.0676 of the mean, across three of five submission slots, before the inversion resolved the question.

Relation to prior work. Recovering labels from leaderboard feedback is established. Blum and Hardt (2015) formalise leaderboard overfitting, Whitehill (2016) inverts an AUC oracle, and Aggarwal et al. (2021) recover labels from log-loss scores, a strictly stronger result than the one reported here. The narrower observations in this paper are that the channel is passive, since the organisers published the baseline row themselves, so no submission is spent and submission-rate defences do not apply; that the leak is in the metric's divisor rather than its numerator; and that it arises in a non-decomposable, non-pairwise metric, where the mechanism is the degenerate confusion structure of a constant predictor. Blum and Hardt identify limiting numerical precision as a poorly understood heuristic; Equation 1 is a concrete case in which four decimals is sufficient to invert.

Level	ST1	ST2	ST3	mean
L1 transcript	0.4916	0.5960	0.4348	0.5074
L2 +video context	0.5251	0.6056	0.4654	0.5321
L3 +channel name	0.5242	0.6057	0.4562	0.5287
L4 +product page	0.6099	0.6993	0.4582	0.5891
paired contrasts, mean macro-F1, 95% CI				
L2 − L1	$+0.0250$	$[+0.0101, +0.0396]$ *		
L3 − L2	-0.0034	$[-0.0125, +0.0045]$		
L4 − L3	$+0.0625$	$[+0.0439, +0.0821]$ *		
L4 − L1, ST3 only	$+0.0237$	$[-0.0094, +0.0579]$		

Table 3: Data access levels, measured on the zero-shot member with only the prompt's field set changed and the decision parameters refitted per level. Paired channel bootstrap, 4,000 replicates, out-of-fold split.

Recommendations. Fixing the label set for the average removes both the variance in §5 and the inversion in one step, at the cost only of rare classes scoring 0 by an explicit rather than an implicit choice. Failing that, the reference baseline can be reported on a split that is not being scored, fewer decimals can be published, or a bootstrapped rather than an exact score can be reported (Chaibub Neto et al., 2016). The adaptive-analysis literature (Dwork et al., 2015) and operational leaderboard design (Lin et al., 2022) bear on the same problem, and the case against reading a leaderboard as a ranking of systems predates this task (Ethayarajh and Jurafsky, 2020). The organisers, to whom this was reported, confirmed the recovery against their own labels (personal communication).

7 What the data and the compute provided

7.1 Data access

The task is built around the question of what a monitoring system achieves at a given level of data access and cost. Because a level is a prompt field set for the zero-shot member, the question can be answered by inference alone. That member was rerun at each level, changing nothing but which fields are serialised, with the decision parameters refitted per level so that each level is scored with thresholds calibrated to its own probabilities (Table 3).

Video context is worth $+0.025$. The channel name adds nothing measurable, with a point estimate of -0.003 and an interval spanning zero. The product page adds the most, $+0.063$, and it is also the only level whose collection the organisers describe as an outbound crawl over pages that may be dead or block automation; the first three levels are metadata calls.

The composition is more informative than the total. The product page's gain falls almost entirely on ST1 ($+0.089$) and ST2 ($+0.096$), the two subtasks concerned with what is being sold. ST3 is flat at every step, $+0.024$ from L1 to L4, with every adjacent interval spanning zero. On this evidence the most expensive level supports product identification and does not support compliance judgement, so a deployment concerned only with the regulatory flags may not require the crawl.

A plausible explanation lies in the labels rather than the model. The organisers report cross-model agreement between their two annotator models of 94.4% exact-label on ST1 but only 44.4% exact-set on ST3. Where two frontier models agree on fewer than half of instances, additional input fields would not be expected to help, and the ST3 ceiling is more plausibly a label ceiling than a data-access one.

7.2 Ensemble anatomy

Table 1 gives the leave-one-out result. The zero-shot 32B model is the most valuable member by a factor of two over the next, with intervals excluding zero on both ST2 and ST3, while consuming 1.5 GPU-hours. Removing DeBERTa-v3-large, six GPU-hours of fine-tuning, changes the mean by -0.0000 with an interval spanning zero. This runs against the LegalLens 2024 finding that top teams *"consistently relied on fine-tuning pre-trained language models"* (Hagag et al., 2024; Tran et al., 2024), and against the broader result that fine-tuned small models outperform zero-shot generative ones on text classification (Bucher and Martini, 2024). Neither is contradicted directly: the measurement here is a marginal contribution inside an ensemble rather than a single-system comparison, and two of the six intervals span zero, so the ordering among the smaller members is not resolved. On this task, however, the prompted member contributed more than the fine-tuned encoders at roughly a quarter of the compute.

7.3 Interventions that did not survive

Of the interventions gated on the paired bootstrap, most were indistinguishable from noise and were dropped. A 16-configuration blend-weight grid moved the out-of-fold mean by $+0.0019$ and moved dev in the opposite direction. A fourth encoder seed, a pure variance-reduction change, moved ST1 down by 0.005. A gradient-boosted ST3 stacker is better standalone (0.6096 against 0.5780 macro over the eight flags) but failed all three integrations into the ensemble, because its errors correlate more closely with those of the other members. Reasoning mode on the zero-shot member moved the system by $+0.0019$, CI $[-0.0011, +0.0071]$. Thresholds chosen to maximise expected per-class F1 at evaluation size promised a large gain on a rare class and delivered -0.0000 on fresh draws, because per-class expected F1 and macro expected F1 are different objectives under present-label semantics. Whether adding legal provisions to the prompt helps was not measured here; Gui et al. (2025) report at this venue that it improves explanation quality without consistently improving detection.

8 The benchmark is channel-disjoint, not advertiser-disjoint

The overview states that brands and destinations are not held out, and the organisers have independently observed the resulting overlap (personal communication). This section quantifies what that is worth. Taking an advertiser to be the registrable domain of the resolved product-page URL, no channel is shared between the evaluation split and the labelled data, but 54.5% of evaluation-split advertisers already appear there, and 70.4% of evaluation rows carry an advertiser that does.

Measured on the development split with training as the only pool, a lookup table that copies an advertiser's majority training label and contains no model at all scores 0.4649 mean macro-F1, against 0.0928 for the majority-class baseline and 0.7352 for the full system. On the 358 development rows whose advertiser is already labelled, it assigns ST1 exactly right 93.0% of the time. The same property holds inside channel-grouped cross-validation, where 66.7% of training rows have their advertiser present in another fold, so out-of-fold estimates are partly measured on advertisers already fitted.

Two qualifications apply. This is not evidence that participating systems perform lookup: the system described here does not, and its development score does not fall on rows whose advertiser was never labelled (0.7126 unseen against 0.6805 seen). The shortcut is also considerably weaker for compliance than for product identity: against the majority baseline the lookup gains $+0.534$ on ST1 and $+0.657$ on ST2 but only $+0.283$ on ST3. That is the same asymmetry the data-level ablation shows from the other direction. What is being sold is substantially memorisable; whether it complies is not. A future edition might report an advertiser-disjoint split alongside the channel-disjoint one.

9 Labels the system did not produce

Four labels in submitted files were set by hand. Three are in the final entry, and one was in an earlier submitted entry and withdrawn before the final one. The complete account is given here because the reconstruction is a single command against the submitted archives, and because the first written account of it, sent to the organisers, was incomplete.

The final entry's three are: one ST1 `none`$\rightarrow$`other` on an Omaze sweepstakes, where the advertiser occurs in two labelled segments carrying gold `other` and `none`; and two ST2 `+gambling` on betting sponsorships, one a PrizePicks read whose advertiser appears in exactly one labelled segment carrying gold `gambling`, and one a Picklebet partnership whose advertiser appears in no labelled segment at all. The fourth, in the earlier entry, converted a genuine `physical_goods` instance to `other` against an advertiser precedent of seven to one; it was a false positive that also removed a `physical_goods` true positive, and it was withdrawn.

All four had been described as resting on advertiser-level precedent. That holds for two of the four, is absent for the Picklebet label, and is contradicted by the fourth. The earlier account also named the wrong advertiser for the Picklebet label. Both corrections were found by diffing the submitted archives rather than by consulting notes, and both were sent to the organisers.

The three labels in the final entry are worth $+0.0723$ of mean macro-F1, of which the single ST1 label accounts for $+0.0676$. That is roughly eight times the 0.0087 separating the top two entries on the final board, and it is large for the reason §5 describes: ST1 `other` has exactly one gold instance in the evaluation split, so a single label constituted the entire gold column for its class and moved one fifth of the ST1 column from zero to full credit. That a single instance carries this weight is a property of the metric rather than of the label.

What the record can establish. These edits were not visible until they were described. The evaluation phase scores an uploaded predictions file and collects a six-field fact sheet; the competition does not require the producing code, so no artefact it retains distinguishes a prediction a system emitted from one entered by hand. Had the same labels been written into the decision layer as a rule keyed on those instances, a rerun would have reproduced the submission exactly, and the sentence volunteered on the fact sheet would have been the only trace. This observation concerns the entry described here only. No evidence is held about any other entry, and the point is that the record does not carry such evidence in either direction. A low-cost remedy is available: collect the producing code, rerun it, diff the output against the submitted file, and inspect any rule that fires on a small number of instances. Such a check would have identified the edits reported here. No check of this kind was run: with no code collected, the provenance of a submitted file was not independently verified, and what is known about it is what teams chose to report.

After the phase closed, the precedent behind the Omaze label was implemented as code, with its brand vocabulary built only from the labelled splits and its parameters fixed before first execution. Applied uniformly to all 503 evaluation instances it selects exactly that one, with no false positives, and the resulting file would return 0.7213. That is reported as evidence that the label was reachable within the rules, and not as an achieved score: it was never submitted, and it was written with prior knowledge of which instance it needed to select.

10 Conclusion

The evaluation of this shared task turned out to be more consequential than any component of the system entered into it. Because macro-F1 was averaged over the labels present in each scored sample, a class with two training instances moved the reported score by several times the gap between the top two entries, made a legitimate modelling decision undecidable from the training data, and exposed the hidden gold counts in closed form from a row the organisers published themselves. On the data-access question the task was built to ask, the answer from this system is that one level adds nothing measurable, the most expensive level supports product identity rather than compliance, and the compliance labels are themselves too unstable for additional input to help. These measurements, rather than a position in the final ordering, are what this paper reports, since by its own noise estimate that ordering cannot resolve the systems it ranks.

Limitations

The data-level ablation is measured on one member of the ensemble rather than on the whole system. The zero-shot lane is the only one in which a level is a pure prompt-field change, so the comparison is clean, but the deployed ensemble contains four other members that read the product page independently, and the system-level marginal value of level 4 is therefore lower than Table 3 indicates. The level-3 null is likewise a property of this system rather than of the level: a field that adds nothing to one architecture can carry signal for another, and whether it does is a cross-system question for the task overview.

The divisor measurement in §5 is made on the out-of-fold split, where gold is available. The inference that the same mechanism operates on the evaluation split rests on the inversion in §6 and on the overview's published counts, since a split whose labels are not held cannot be bootstrapped directly.

The leave-one-out figures in Table 1 refit the decision layer per arm and are therefore marginal contributions within one specific ensemble rather than measures of intrinsic model quality. Two of the six intervals span zero.

Roughly sixteen changes were gated at a 95% level without a multiplicity correction, and beneath those sit more than two thousand uncorrected threshold-grid evaluations. The accepted changes should be read accordingly: a family-wise correction across the gated tests leaves fewer of them significant than their individual intervals suggest.

The advertiser analysis uses the resolved product-page domain as a proxy for advertiser identity. It will merge distinct advertisers that share a platform domain, and will split advertisers across country domains.

Ethical Considerations

The predictions submitted were not produced entirely by the system described here, which conflicts with the competition Terms; §9 states this in full. The matter was raised with the organisers after the evaluation phase closed and before any external party had identified it, with the affected instances and the corrected accounting supplied, and with an undertaking to accept any adjustment they considered appropriate. The system-only score is reported throughout and no rank is claimed.

The dataset concerns commercial content on channels likely to reach children. The analysis here operates at the level of the benchmark and the metric. Advertisers are identified only where they already appear in the released labelled data and where doing so is necessary to substantiate a claim, and no statement is made about any individual creator's compliance. Outputs of this kind identify candidates for review rather than legal findings, as the organisers themselves note.

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A Reproducibility

Exact models. `microsoft/deberta-v3-large`; `answerdotai/ModernBERT-large`; `Qwen/Qwen3-32B` (zero-shot, served offline by vLLM); `Qwen/Qwen3-14B` (LoRA sequence classifier).

Encoders (`m1d`, `m1m`). Three seeds each $(42, 43, 44)$, 5 sequential folds per run, one shared trunk with three heads (softmax ST1, sigmoid ST2 and ST3), AdamW, trunk lr 10^{-5}, head lr 5×10^{-5}, weight decay 0.01, batch 16, 8 epochs. Inputs are assembled under per-field token budgets: DeBERTa (`max_len` 512) transcript 290, title 22, description 90, channel 12, page title 14, page text 60; ModernBERT (`max_len` 2048) 1100 / 32 / 220 / 16 / 32 / 560. The transcript is placed first so that tail truncation consumes the page text rather than the segment. Non-transcript fields are dropped independently at $p = 0.15$ during training, seeded per run and fold. Member probabilities are the mean over seeds.

Zero-shot (`m3q32A`). Temperature 0, fixed seed, guided JSON decoding against a per-task schema, top-k logprobs retained. The per-label probability is taken from the logprobs of the `yes`/`no` value token rather than from the decoded string, with a text-parse fallback. The system prompt quotes the task taxonomy verbatim and prepends a data-guard instruction; the user prompt serialises the instance under the level-4 field set, which is the only element varied in Table 3.

LoRA (`m2qS2`). $r = 32$, $\alpha = 64$, dropout 0.05, all linear projections, bf16, `max_len` 2048, LoRA lr 10^{-4} and head lr 2×10^{-5}, 4 epochs, effective batch 8, one run per sub-task over the same 5 folds. Only the ST2 head is used in the final system.

Classical and stacking lanes. `m0s`: three word (1–2)-gram TF-IDF blocks over transcript, title-plus-description and page text at 40k features each, plus one character-`wb` (3–5)-gram block at 50k, `min_df` 2, sublinear tf, one-vs-rest logistic regression, $C = 4.0$, balanced class weights. `s3s`: logistic regression, $C = 1.0$, balanced class weights, over the fold-honest member probabilities plus ten auxiliary features, comprising three disclosure-state indicators, log word counts of transcript, description and page text, and four sponsorship and disclosure regex hits on the transcript and description. Stacker folds reuse the channel-grouped assignment, so no instance is stacked on a model that saw its channel.

What was fitted on which split. Folds are a 5-way `GroupKFold` over the 632 training channels with greedy rare-label balancing. Every member is trained within folds. Every threshold, the ST1 prior and τ, and the four ST3 cascade parameters are fitted on pooled out-of-fold predictions. The development split was used as a direction check and as a veto on two candidate changes, which is a mild form of selection on dev; the evaluation split was never used for fitting. Per-class thresholds are chosen by a grid over $[0.01, 0.99]$ maximising out-of-fold F1, with two guards: classes with fewer than 15 positives fall back to a plug-in value, and classes whose per-fold optimum spreads by more than 0.35 are floored at half the achieved F1.

Code. All measurements in this paper are produced by scripts in the repository at https://github.com/thylinao1/childsafeads: `clinic/presence_artefact.py` (§5), `clinic/level_ablation.py` (Table 3), `clinic/member_ablation.py` (Table 1), `clinic/advertiser_overlap.py` (§8) and `clinic/hand_label_audit.py` (§9), with the pinned scorer in `eval/local_scorer.py`.